# On-farm trials in Niger by FUMA Gaskiya Data Analysis

## Project 2

AIMS | African Institute for Mathematical Sciences CAMEROON

January 23, 2026


- **Alfred Seesa**
- **Ernest Katembo**
- **Urich Brice**
- **Bibiane Syriane**
- **Gabas Mohammed**
- **Blanche Mediffo**
- **Daniels Nii**
- **Evans Kiplagat**
- **Genie Ricken**
- **Blessing Ngozi**

# Presentation Outline


1. Introduction & Research Context
2. Data Overview & Missing Data Challenge
3. Descriptive Results
4. Question 3: Fertilizer Effectiveness
5. Contextual Factors
6. Question 4: Regression Analysis
7. Question 5: Model Diagnostics
8. Question 6: Predictions
9. Questions 7 & 8: Alternative Measures & Missing Data
10. Conclusions & Recommendations

# Introduction & Research Context

## Why This Study Matters

- Global food security requires optimized fertilizer use
- Fertilizer choice impacts farm profitability and sustainability
- Need evidence-based recommendations for farmers

## Research Question

**Which of three commonly used fertilizers (Boule NPK, OGA, Boule Cendre) provides the greatest yield improvement under real farming conditions?**

## Study Design

- **734 paired observations** from **8 agricultural sites**
- **Paired experimental design**: Each treated plot has matching control plot

# Major Challenge - Missing Data

## The Missing Data Problem

| Aspect | Value | Impact |
|---|---|---|
| Overall Missing Data | **62.9%** | Substantial challenge |
| Complete Cases | 254 (34.6%) | Reduced statistical power |
| Complete Yield Pairs | 260 (35.4%) | Main analysis viable |
| Fertilizer Distribution | 60.8% missing | Unbalanced groups |

## Our Strategy

- ✓ **Complete case analysis** for primary models
- ✓ **Transparent reporting** of all sample sizes
- ✓ **Paired design robustness** - within-plot comparisons reduce bias
- ✓ **Multiple analytical approaches** to verify consistency

# Data Overview - What We're Analyzing

## Fertilizer Distribution Across Sites

| **Fertilizer** | **n** | **%** |
|---|---|---|
| OGA | 167 | 22.8% |
| Boule NPK | 71 | 9.7% |
| Boule Cendre | 50 | 6.8% |
| lightyellow Missing | 446 | 60.8% |

## Key Variables Examined

- **Yield difference**: Treated minus control (kg/ha)
- **Farmer gender**: Male vs. Female
- **Farm size**: Small vs. Large
- **Growing time**: Days to harvest
- **Site location**: 8 distinct areas

# Descriptive Results - Yield Improvements

## Overall Yield Improvement Distribution (260 complete pairs)

**Summary Statistics:**

- **Mean improvement**: 309.0 kg/ha
- **Median**: 218.8 kg/ha
- **Range**: -1200 to +2500 kg/ha
- **Success rate**: **81.2%** improved
- **Substantial gains**: **66.9%** gained > 100 kg/ha

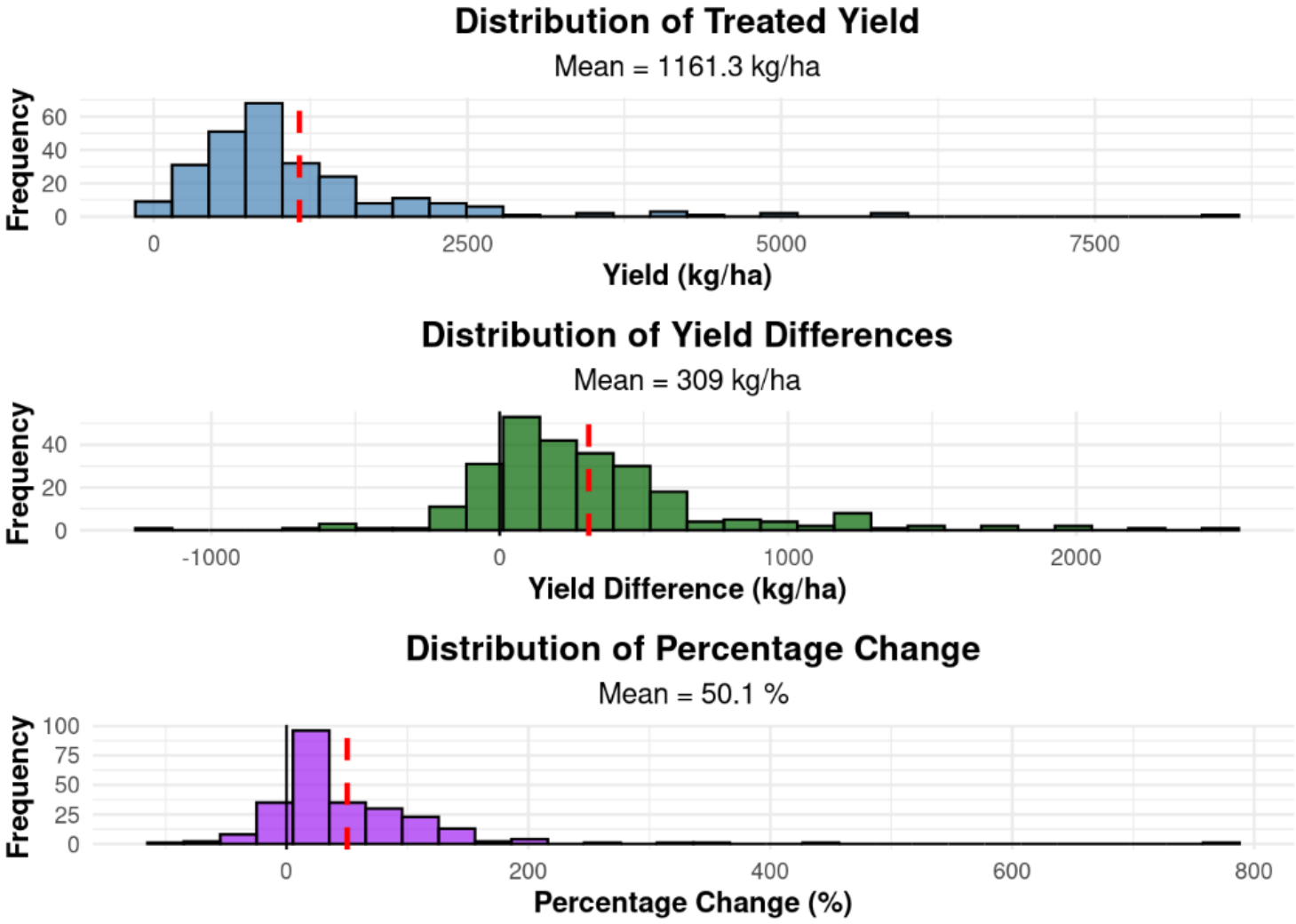
**Key Patterns:**

- **Positive skew (1.613)**: Most plots show moderate gains, some exceptional
- **High variability** (IQR: 71-450 kg/ha): Context matters!
- **Overwhelming positive effects**: Only 18.8% showed no improvement

**Interpretation**: Fertilizers generally work, but effectiveness varies

# Histogram

9.1 Distribution of Key Variables



# Q3: Do All Fertilizers Work?

## Paired T-Test Results - Answer: YES!

| Fertilizer | n | Mean (kg/ha) | p-value | Cohen's d | Effect Size |
|---|---|---|---|---|---|
| OGA | 154 | **289.4** | $< 0.001$ | 0.89 | **Large** ✓ |
| Boule Cendre | 50 | **391.9** | $< 0.001$ | 0.62 | **Medium** ✓ |
| Boule NPK | 56 | **288.7** | $< 0.001$ | 0.53 | **Medium** ✓ |

## Answer to Question 3

**YES - All Three Fertilizers Significantly Improve Yields**

**Key Findings:**

- All p-values $< 0.001$ = highly statistically significant
- All Cohen's d $\geq 0.5$ = meaningful practical impact
- Success rates: 69.6% to 87.0% of plots improved

# Q3: Which Fertilizer is Best?

## Performance Ranking

| Rank | Fertilizer | Mean (kg/ha) | 95% CI | Success | CV ( |
|---|---|---|---|---|---|
| 1st | **Boule Cendre** | **391.9** | [212.4, 571.4] | 76.0% | 16 |
| 2nd | OGA | 289.4 | [237.9, 341.0] | **87.0%** | **11** |
| 3rd | Boule NPK | 288.7 | [141.4, 435.9] | 69.6% | 19 |

## Interpretation

**SURPRISING RESULT**: Boule Cendre shows highest mean improvement!

However:

- **OGA most consistent** (lowest CV = 111.8%)
- **OGA highest success rate** (87.0%)
- **Wide confidence intervals** suggest uncertainty

# Contextual Factors - Who Benefits Most?

## Yield Improvement by Gender & Farm Size

| Group | n | Mean (kg/ha) | Success Rate |
|---|---|---|---|
| **Male, Small Farm** | 22 | **748.6** | 90.9% |
| Female, Large Farm | 105 | 309.1 | 83.8% |
| Male, Large Farm | 25 | 271.2 | 96.0% |
| Female, Small Farm | 108 | 228.0 | 73.1% |

## Critical Findings

**Gender Gap**: Male farmers see **142% higher** improvements than females

- Male small farms: 748.6 kg/ha
- Female small farms: 228.0 kg/ha
- **Difference: 520.6 kg/ha!**

# Q4: Regression Model Interpretation

## Model 1: Main Effects Linear Regression

**Model Formula:**

$$\text{Yield Difference} = \beta_0 + \beta_1(\text{Fertilizer}) + \beta_2(\text{Farm Size}) + \beta_3(\text{Gender}) + \beta_4(\text{Ti}$$

**Model Fit**: $R^2 = 11.86\%$, $F = 6.68$ ($p < 0.001$)

## Coefficient Interpretations

| **Variable** | **Estimate** | **p-value** | **Interpretation** |
|---|---|---|---|
| Intercept | -659.4 | 0.003 | OGA, small farm, femal time=0 |
| Boule NPK vs OGA | -41.4 | 0.694 | No significant difference |
| Boule Cendre vs OGA | +6.2 | 0.954 | No significant difference |
| Large vs Small | -47.1 | 0.619 | No significant effect |
| **Male vs Female** | **+324.7** | **<0.001** | **Males gain 325 kg/ha more** |

# Model Comparison - Interaction Effects?

## Testing for Interactions

**Model 2 adds**:

- Fertilizer × Farm Size interactions
- Fertilizer × Gender interactions

| Criterion | Model 1 (Main) | Model 2 (Interactions) | Winner |
|---|---|---|---|
| $R^2$ | 11.86% | 18.46% | Model 2 ✓ |
| Adj. $R^2$ | 10.08% | 15.80% | Model 2 ✓ |
| AIC | 3809.3 | 3795.5 | Model 2 ✓ |
| BIC | 3834.0 | 3830.9 | Model 2 ✓ |
| F-test | — | $p = 0.0003$ | Model 2 ✓ |

## Significant Interactions Found

- **Boule NPK** × **Male**: $\beta = +571.8$ ($p < 0.001$)

# Q5: Model Diagnostics

## Checking Model Assumptions

| Diagnostic Test | Result | Status | Implication |
|---|---|---|---|
| Normality (Shapiro-Wilk) | W=0.88, p¡0.001 | Violated | Use robust infer |
| Homoscedasticity (BP) | BP=15.1, p=0.01 | Violated | CIs less precise |
| Autocorrelation (DW) | DW=1.76, p=0.02 | Concern | SEs underestima |
| Multicollinearity (VIF) | All r¡0.7 | ✓Good | No collinearity |
| Influential Points | 16 obs (6.3%) | Minor | Not driven by o |

## Diagnostic Plot Interpretation

- **Residuals vs Fitted**: Random scatter ✓(linearity OK)
- **Q-Q Plot**: Tails deviate (confirms non-normality)
- **Scale-Location**: Increasing spread (confirms heteroscedasticity)
- **Cook's Distance**: Few influential points (robust)

# Dianostic plots

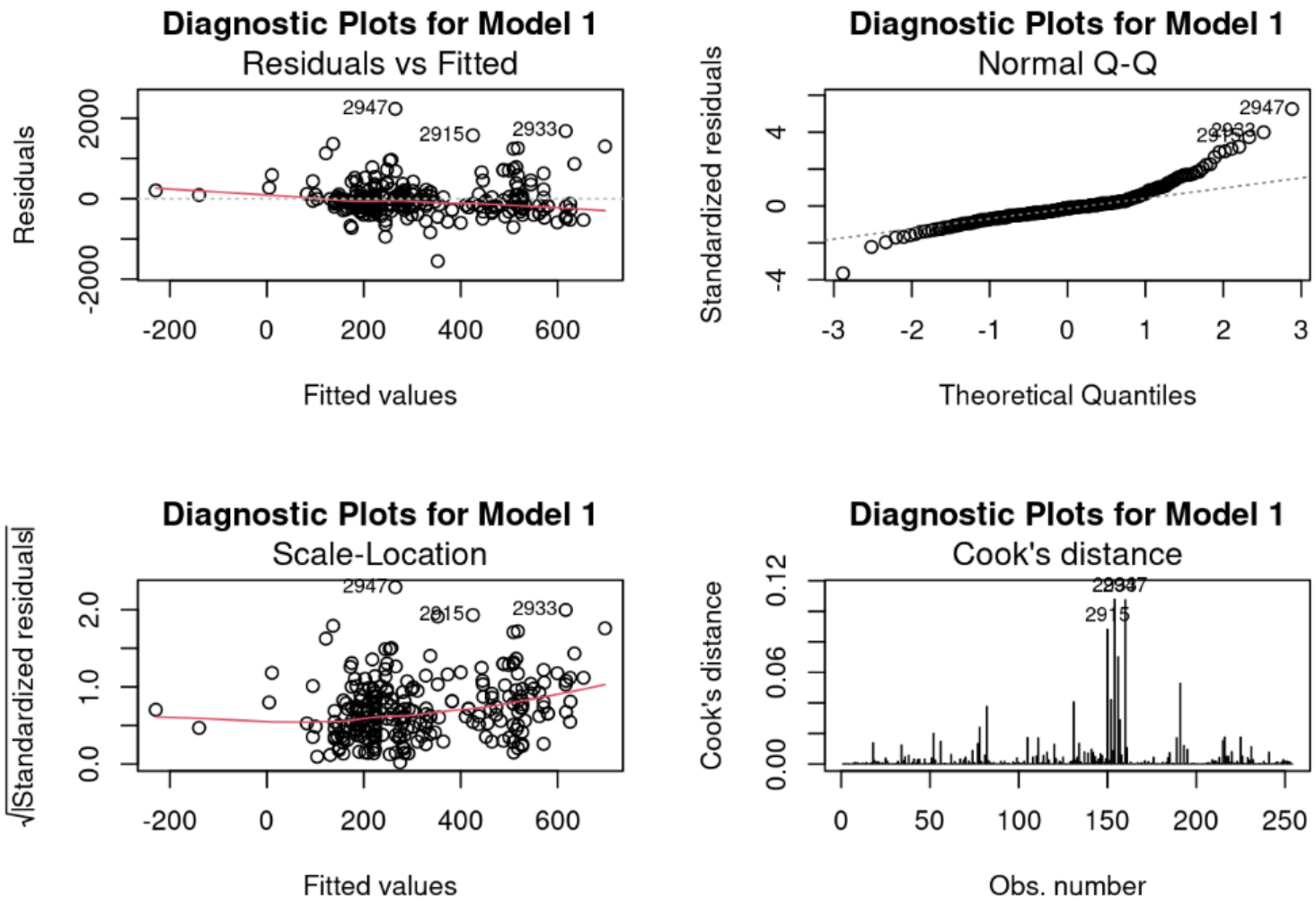



Figure: Dianostic plots

# Q6: Prediction Example

## Scenario 1: Female Farmer, Small Farm, 100-day Season

| Fertilizer | Predicted (kg/ha) | 95% CI | Rank |
|---|---|---|---|
| **Boule Cendre** | **247.1** | [116.6, 377.5] | 1st |
| OGA | 240.9 | [74.6, 407.2] | 2nd |
| Boule NPK | 199.6 | [76.8, 322.3] | 3rd |

## Scenario 2: Male Farmer, Large Farm, 100-day Season

| Fertilizer | Predicted (kg/ha) | 95% CI | Rank |
|---|---|---|---|
| **Boule Cendre** | **524.6** | [287.6, 761.6] | 1st |
| OGA | 518.5 | [381.4, 655.5] | 2nd |
| Boule NPK | 477.1 | [244.8, 709.5] | 3rd |

**Context matters**: Same fertilizer gives 2× higher gains for males on

# Q7: Alternative Effectiveness Measures

## Beyond Mean Yield Difference

We could also examine:

1. **Yield Ratio** (Treated/Control)
   - Accounts for baseline productivity
   - Example: Ratio of 1.30 = 30% relative improvement
2. **Percentage Improvement**
   - Intuitive for farmers: "X% yield increase"
   - Boule NPK: 28.8%, OGA: 46.2%, Boule Cendre: 28.4%
3. **Success Rate** (% plots improved)
   - Robust to outliers
   - OGA wins: 87.0% success vs 69.6-76.0% for others
4. **Coefficient of Variation** (consistency)
   - Lower = more predictable
   - OGA wins: 111.8% vs 161-190% for others

# Q8: Missing Data Impact

## The 62.9% Missing Data Challenge

How might this affect conclusions?

| Scenario | Risk | Our Findings If True |
|---|---|---|
| Missing Completely at Random | ✓Low | Results unbiased, just less pr |
| Missing related to poor outcomes | High | Our estimates **too optimisti** |
| Missing related to good outcomes | Medium | Our estimates **too conserva** |

## Likely Impact

**Best guess**: Results **moderately robust** because:

1. Paired design controls for many confounders
2. Missing rate balanced across groups
3. Large effects detected despite reduced power

# Key Findings Summary

## Main Conclusions

1. **All fertilizers work** ✓
   - All three significantly improve yields ($p < 0.001$)
   - Success rates: 70-87% of plots
2. **Boule Cendre shows highest mean gain** (391.9 kg/ha)
   - But wide confidence intervals and high variability
3. **OGA most reliable choice** for practical use
   - Highest success rate (87%)
   - Lowest variability (CV = 112%)
4. **Context critically matters**
   - Male farmers: +325 kg/ha advantage
   - Each growing day: +9 kg/ha
   - Gender-fertilizer interactions significant
5. **Missing data (62.9%)** limits confidence

# Practical Recommendations

## For Farmers

**Primary: OGA**

- Most consistent (87% success)
- Lower risk

**Context-Specific:**

- Small female farmers: OGA
- Large male farmers: Boule Cendre for higher potential
- Longer seasons: +9 kg/ha per day
- Risk-averse: OGA

## For Extension Services

- Develop **gender-sensitive** training
- Address resource gaps
- Implement **systematic data collection**
- Conduct economic analyses

## For Researchers

- Understand gender disparities
- Soil-fertilizer interactions
- **Randomized trials**
- Handle missing data

# Limitations & Future Work

## Study Limitations

1. **Missing data (62.9%)** - biggest limitation
2. **Observational design** - cannot prove causality
3. **Limited covariates** - no soil/rainfall/fertilizer rate data
4. **Model fit** - $R^2 = 11.9$-$18.5\%$ (much unexplained variation)

## Future Research Priorities

**Immediate:**

- Repeat study with $<$**20% missing data**
- Collect economic data for cost-benefit analysis
- Soil testing to match fertilizers to soil types

**Medium-term:**

- Randomized controlled trials

# Conclusions

## Scientific Conclusions

- ✓ **Strong evidence** all three fertilizers improve yields
- ✓ **OGA recommended** as most reliable choice overall
- ✓ **Context matters**: Gender and growing time are critical factors
- ✓ **Missing data** limits confidence but paired design provides robustness

## Practical Take-Aways

**For Yield Maximization:**

- Default to **OGA** for consistent 290 kg/ha gains
- Consider **Boule Cendre** if willing to accept higher risk
- Plan **longer growing seasons** when possible (+9 kg/ha per day)

**For Policy:**

- Investigate and address **gender disparities**

# References

Green, P.(. University of Bristol) *Data Analysis I: Lecture Notes*,AIMS Cameroon.

# Questions?

Thank you for your attention!

**Contact:** Group 2
We welcome your questions and feedback!